

# ADITI CHOWDHURI

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## EDUCATION

**Columbia University, New York, NY** Master of Science in Computer Science - **CGPA: 3.91/4** 2024 – 2025  
Relevant Coursework: Advanced Software Engineering, Analysis of Algorithms, User Interface Design. Honors: TAI for 4111 (Introduction to Databases) and 4223 (Networks Crowds and the Web)

**VIT, Vellore, IN** Bachelor of Science in Computer Science and Engineering - **CGPA: 3.99/4** 2018 – 2022  
Relevant Coursework: Database, Web Programming, Data Visualization. Honors: Special Achievers and Achievers 2021

## SKILLS

- **Programming Languages:** Python, Java, C, C++, C#, R, MATLAB, SQL, JavaScript, Go
- **Frameworks:** TensorFlow, Keras, PyTorch, Numpy, Pandas, Matplotlib, Plotly, OpenCV, Beautiful Soup, Gunicorn
- **Tools:** Git, Arduino, Raspberry Pi, Figma, Adobe Photoshop, Google Firebase, Google Firestore, GCP, AWS, Linux
- **Web Development:** HTML5, CSS3, JQuery, MongoDB, Flask, NodeJS, ExpressJS, ReactJS, D3 JS, EJS, JSX

## EXPERIENCE

**Research Intern** Sep 2024 - Dec 2024  
ARISE Lab - Columbia University *New York, US*

- Creating AI solutions to enhance LLMs with improved semantic understanding for better security.

**Software Engineer** June 2022 - July 2024  
Bank of America *Mumbai, IN*

- Engineered and deployed user-centric UI features and backend services for financial applications, significantly enhancing user experience and increasing system performance by 20%.
- Developed machine learning models focused on data preprocessing and Natural Language Processing (NLP), improving data accuracy and processing efficiency by 25%.
- Implemented continuous integration and deployment pipelines, reducing deployment time by 40% and improving code quality.

**Software Developer Intern** July 2021 - Oct 2021  
Tata Consultancy Services *Mumbai, IN*

- Designed, developed, and maintained scalable microservices using Spring Boot and Java to support enterprise applications.
- Worked with MySQL and PostgreSQL for efficient data management and query performance tuning.
- Optimized application performance by refactoring code, reducing runtime by 10%, and improving overall efficiency.

**Web Developer Intern** Aug 2020 - Dec 2020  
Trikon Technologies *Noida, IN*

- Designed, developed, and maintained the company website, enhancing user experience and responsiveness.
- Collaborated with the backend team to integrate RESTful APIs using Node.js and Express.js, improving data retrieval efficiency by 30%.
- Conducted website testing and debugging to resolve performance issues and enhance overall stability.

## PROJECTS

- **Foodgasm:** Developed "FoodGasm," an AI-powered recipe platform using computer vision to detect fridge ingredients and LLMs for personalized recipe suggestions. Built with React.js, FastAPI, Microsoft Phi 3.5 Vision Instruct, and AI-driven calorie estimation for a seamless and intelligent user experience.
- **Tutor AI:** Developed an AI-based posture correction system using PoseNet, TensorFlow, and JavaScript to provide real-time feedback for physically demanding tasks. Integrated machine learning to analyze body alignment and offer personalized corrections. Used React.js for the user interface, ensuring a seamless and interactive learning experience.

## RESEARCH PAPERS

- **Temporal Weighted Averaging for Asynchronous Federated Learning**  
Computational Intelligence and Neuroscience 2021, Hindawi - DOI: [10.1155/2021/5844728](https://doi.org/10.1155/2021/5844728)
- **Progressive Search Personalization and Privacy Protection Using Federated Learning**  
Expert Systems, Wiley - DOI: [10.1111/exsy.13318](https://doi.org/10.1111/exsy.13318)